II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 11

Stones, Obstruction & Trauma

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · **P2** = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 93	Management of Upper Urinary Tract Obstruction
Ch 94	Upper Urinary Tract Trauma
Ch 95	Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis
Ch 96	Acute Care of Urinary Lithiasis
Ch 97	Surgical Management of Ureteral Calculi
Ch 98	Surgical Management of Upper Urinary Tract Calculi
Ch 99	Prevention of Urinary Stone Disease
Ch 100	Lower Urinary Tract Calculi
Ch 54	Management of Paediatric Kidney Stone Disease
Ch 125	Genital and Lower Urinary Tract Trauma
Ch 62	Paediatric Genitourinary Trauma
★	Putting it together

93 Management of Upper Urinary Tract Obstruction

1 · URETEROPELVIC JUNCTION OBSTRUCTION (UPJO): PRESENTATION

- Trauma: see Ch 94, Ch 125; stones: see Ch 95–100.
- UPJO is usually congenital. **Neonates** show prenatal hydronephrosis on ultrasonography, with azotaemia if bilateral or solitary.
- **Older children and adults** have intermittent flank pain with vomiting, haematuria after minor trauma, urinary tract infection (UTI) and, rarely, hypertension.

2 · DIAGNOSIS

Prove impaired drainage before operating.

> **KEY POINT**
>
> **Dilation is not obstruction.** Anatomical imaging (ultrasonography, computed tomography [CT], retrograde) shows where; functional tests (renography, Whitaker) show whether it matters. Combine one of each.

Modality	Key point
Ultrasonography (first in neonates)	Large medial pelvis, rounded calyces joined by dilated infundibula (unlike multicystic kidney); ureter not seen
Non-contrast CT (acute pain)	Hydronephrosis to UPJ without stone; excludes radiolucent stone, tumour
Contrast CT	Shows **crossing vessels**; can substitute for renography
Diuretic renography	Differential function, drainage; best follow-up study

Modality	Key point
Retrograde pyelography	**At the time of repair** (avoids introducing infection)
Nephrostomy and antegrade study	When retrograde fails; tests recoverability; permits Whitaker

P1 *Radionuclide studies and diuretic renography in PUJ obstruction — tracers, well-tempered protocol, T½ cut-offs, P40, O'Reilly curves*

- **Renography**: hydrate, then give furosemide **0.3–0.5 mg/kg** at **20 min**; mercaptoacetyltriglycine (MAG3) is preferred.
- **Obstruction is a half-time (T½) >40 min**. After successful pyeloplasty, drainage occurs even before furosemide.

ALGORITHM

1. **Whitaker test**
2. Place a nephrostomy and a bladder catheter; perfuse at **10 mL/min**.
3. If pressure stays ≤12–15 cm H2O, there is no obstruction; if it exceeds **15–20 cm H2O**, there is.
4. It is invasive, so use it only when other tests are non-diagnostic.

3 · INDICATIONS AND CHOICE OF INTERVENTION

- **Indications**: symptoms; impaired or falling function; stones or infection; rarely hypertension.
- **Neonates** are selectively observed; operate for progressive hydronephrosis or a **10%** fall in differential function. **Asymptomatic adults** with indeterminate results have serial renography.
- The **procedure of choice** is dismembered pyeloplasty, now robotic at most major centres; endopyelotomy fails more.
- Drain before repair only for infection, azotaemia (solitary or bilateral) or unrelenting pain.

Endopyelotomy subgroup (landmark series)	Success
Crossing vessel present vs absent	**42% vs 86%**
High- vs low-grade hydronephrosis	60% vs 81%
Both excluded	95% (≈ pyeloplasty)

- A crossing vessel is **prognostic, not a contraindication**, but it pushes primary UPJO towards pyeloplasty.
- **Secondary UPJO: cross over.** Failed endopyelotomy is treated by pyeloplasty; failed pyeloplasty by endourology first.
- **Nephrectomy** suits a thin shell with a normal contralateral kidney, stones with chronic infection, or multiple failed repairs. Adults with **<15%** differential function are non-salvageable; if uncertain, stent or drain and re-scan.

4 · ENDOPYELOTOMY

- **Principle**: a full-thickness **lateral** incision into peripelvic fat heals over a stent; lateral is **free of crossing vessels**.
- Pyeloplasty (≈**95%**) wins because it makes a wide dependent funnel; endourology wins only on recovery.
- **Contraindications**: segment >**2 cm**, active infection, uncorrected coagulopathy. Urine must be sterile first.

Percutaneous antegrade

- Best for **UPJO with pelvicalyceal stones**. Puncture the **mid-posterior or superolateral calyx**, not the lower pole, whose acute infundibular angle blocks passage.
- Stent with a 14/7-Fr endopyelotomy stent, traditionally for **6 weeks**; 2 and 4 weeks are equal.

- **Clear stones before incising**, because fragments migrate periureterally and cause fibrosis.

Retrograde ureteroscopic (standard)

P1 *Retrograde endopyelotomy — indications, holmium lateral-incision steps, results (≈90% without crossing vessel >4 mm), complications, balloon/cautery-wire alternatives*

Not in the paper answer:

- Control bleeding with haemostatic laser settings or 10-min balloon tamponade.

ALGORITHM

1. General anaesthesia; working and safety wires; flexible ureteroscope via a sheath.
2. With a 200-µm holmium fibre, incise **laterally** into fat and into normal ureter.
3. Leave a stent and Foley; do a renogram 4 weeks after stent removal.

- Percutaneous endopyeloplasty (endoscopic Heineke–Mikulicz) fails in secondary UPJO.

Complications and follow-up

ALGORITHM

1. **Percutaneous bleeding**
2. Bed rest and transfusion.
3. **Do not irrigate** the nephrostomy, so clot tamponades.
4. If bleeding persists, do **selective embolisation**.

- Renogram at 1 and 6 months, then yearly for **2–3 years**. Pyeloplasty 85/75% vs endopyelotomy 63/41% at 3/10 years.
- After failure, get CT angiography; a significant crossing vessel needs **pyeloplasty, not repeat incision**.

5 · PYELOPLASTY

KEY POINT

Principles (open = minimally invasive): widely patent, watertight, tension-free anastomosis; funnel-shaped, **dependent** transition.

P3 P1 *Laparoscopic pyeloplasty; Management of failed pyeloplasty — Anderson–Hynes steps, approaches, contraindications, stent 4–6 weeks, vascular hitch, late failures, redo options*

Not in the paper answer:

- Dismembered (Anderson–Hynes 1949) repair suits any insertion, allows pelvic reduction and **transposition over crossing vessels**, and alone excises the abnormal UPJ.
- Open approach is flank extraperitoneal by default; anterior for horseshoe or pelvic kidney.

ALGORITHM

1. Dissect the ureter with periureteral tissue; place a lateral marking stitch.
2. Excise the UPJ; spatulate **laterally**; bring the apex to the inferior pelvis, anterior to any crossing vessel.
3. Do a watertight absorbable anastomosis, with or without reduction; leave a stent and drain.

- Remove the **stent** at **4–6 weeks**. Reserve an added nephrostomy for secondary UPJO.
- Laparoscopic and robotic success is **>90–95%**, equal to open; the robot's advantage is ergonomic.

ALGORITHM

1. Remove the Foley at 24–36 h, then the drain if output is negligible.
2. If output rises after Foley removal (reflux along the stent), **replace the Foley for 7 days**.
3. Remove the stent at 4–6 weeks; then renography.

6 · OTHER UPJ RECONSTRUCTIONS

> **KEY POINT**
>
> Every UPJ repair asks **what tissue bridges the gap while keeping blood supply**. Dismemberment excises disease but devascularises; flaps keep blood supply but carry disease forward; the calyx is used when the pelvis is too small.

Procedure	Indication	Note
Foley Y-V-plasty	**High insertion**	Not with crossing vessels or need for reduction
Culp–DeWeerd spiral flap	Large pelvis, long proximal narrowing	Longest flap; length:width ≤**3:1**
Scardino–Prince vertical flap	Dependent UPJ, box-shaped pelvis	Shorter flap
Davis intubated ureterotomy	Long strictures, with spiral flap	Heals over stent (secondary intention)
Ureterocalicostomy	Small intrarenal pelvis, horseshoe, failed pyeloplasty	Ureter to lower-pole calyx

- Outline flaps longer than needed; trim the apex, never the base.

P2 *Uretero-calycostomy — resect lower-pole parenchyma (not nephrotomy), mucosa-to-mucosa, omental cover; cortex <5 mm predicts failure*

7 · RETROCAVAL URETER

P3 *Retrocaval ureter — venous anomaly, always right, fish-hook/S-shape, CT; transpose ureter anterior to IVC*

8 · URETERAL STRICTURE — EVALUATION AND ENDOUROLOGY

Exclude cancer; endoscopy suits only short, well-vascularised strictures.

- **Causes**: malignancy, stone, radiation, surgical ischaemia, periureteral fibrosis, endoscopy, infection (tuberculosis, schistosomiasis).
- **Exclude malignancy first** with CT, with or without ureteroscopy and biopsy.
- Stricture follows ureteroscopy (URS) in ≈**3%** vs 1.5% after shock-wave lithotripsy; an impacted stone >2 months gives ≈1 in 4.
- Endourology needs ipsilateral function ≥**25%** (the algorithm uses 20%).
- **Stents** suit intrinsic strictures; extrinsic compression needs tandem or metallic stents. Chronic stenting suits unfit or cancer patients.
- **Balloon dilation** is inflated until the **waist disappears**, then stented 2–4 weeks. Contraindications: infection, stricture >**2 cm**. It works best (≈85%) in short, recent, non-anastomotic, well-vascularised strictures; anastomotic ≈50%.
- **Endoureterotomy** (retrograde holmium) is first line under 2 cm. Cut full thickness into **2–3 mm of normal ureter** each side, over a safety wire.

Level	Incision	Avoids
Distal/mid ureter	**Anteromedial**	Iliac vessels
Proximal ureter	**Lateral/posterolateral**	Great vessels

- Complete obliteration in high-risk patients can be crossed by **cutting toward the light**, then stenting 8–10 weeks.

ALGORITHM

1. If <2 cm and function >20%, then endoscopic incision.
2. If >2 cm and function >20%, then reconstruction.
3. If function <20%, then nephrectomy (children may still merit repair).
4. Transplant ureter: consider balloon. Follow with renography for 2 years.

KEY POINT

Each failure predictor (long, old, anastomotic, poorly vascularised) is a proxy for **ischaemic scar** that an incision cannot heal across.

9 · URETERAL STRICTURE — RECONSTRUCTION LADDER

Defect length and level choose the repair.

Technique	Defect bridged	Level
Ureteroureterostomy	**2–3 cm**	Upper/mid
Ureteroneocystostomy	4–5 cm	Distal
Psoas hitch	**6–10 cm**	Below pelvic brim
Boari flap	**10–15 cm**	Lower/mid; spiral may reach pelvis
Renal descensus	5–8 cm	Upper; adds to any repair

- **Ureteroureterostomy** suits short defects only, because tension causes stricture. Preserve adventitia, spatulate 180° apart, stent and cover with omentum; success is >90%.

ALGORITHM

1. **Persistent leak**
2. X-ray the stent position.
3. Pull the drain off the anastomosis and stop suction.
4. Place a Foley and give an anticholinergic; if it persists, place a nephrostomy.

P3 P2 *Boari flap; Trans Uretero Ureteric anastomoses — contracted bladder contraindicates; recurrent stricture commonest; eGFR fell in 44%; TUU donor above IMA, contraindications, correct recipient reflux*

- In adults, refluxing and non-refluxing reimplants are equal. Treat outlet obstruction first.
- **Renal descensus** rotates the kidney inferomedially and fixes its lower pole; the renal vein limits descent.
- **Ileal ureter** is isoperistaltic. Contraindications: creatinine **>2 mg/dL** (acidosis in ≈half), bladder dysfunction, inflammatory bowel disease, radiation enteritis. Start endoscopic surveillance from year 3.
- **Buccal mucosa onlay** suits dense scar. **Autotransplantation** suits an absent or poor contralateral kidney, or when other repairs are impossible.

KEY POINT

Ladder: bring the bladder up, the kidney down, borrow the other ureter, or substitute (ileum, buccal). Tension and ischaemia cause failure at every rung.

10 · URETEROENTERIC ANASTOMOTIC STRICTURE

- Strictures occur in 4–9% of conduits, so **upper-tract surveillance is lifelong**.
- **The cause is ischaemia**. The **left is commoner**, because it needs extra length, passes under the sigmoid mesentery and angulates at the inferior mesenteric artery.
- Risks: **non-refluxing anastomosis** (≈13% vs 2–3%), carcinoma in situ at the margin, obesity and major complications.

- Conduit dilation is often non-obstructive, so confirm with renography; rapid progression with a mass suggests recurrent cancer.
- **Open revision is the gold standard** (≈91% vs 26% endoscopic); robotic is equal. Endourology is **antegrade**. Endoscopic success is ≈50% under 1 cm and better on the right.

ALGORITHM

1. If <2 cm and function >20%: right side, endoscopic incision; **left side, formal repair** (sigmoid mesentery bleeding).
2. If >2 cm, then formal repair.
3. If function <20%, then nephrectomy or observation.

11 · RETROPERITONEAL FIBROSIS

P3 P1 *Retroperitoneal fibrosis — idiopathic 70% vs secondary (drugs, lymphoma); plaque at L4–L5, medial deviation; PET for activity; deep biopsies; drain, then prednisone (80% respond), then bilateral ureterolysis and omental wrap*

ALGORITHM

1. If uraemic, decompress with a **nephrostomy** or stents and manage the **post-obstructive diuresis**.
2. Stop any culprit drug; biopsy.
3. Give steroids; if they fail, do ureterolysis.

- Steroids work while the lesion is cellular; second line is immunosuppressants or **tamoxifen**. Ureterolysis with an omental wrap is most definitive.

94 Upper Urinary Tract Trauma

P2 *Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management — mechanism, haematuria, AAST 2018, imaging indications, CT red flags, non-operative care, exploration, vascular control, renovascular injury, complications*

Not in the paper answer:

- Polar injury is treated by partial nephrectomy with omental cover.

1 · RENAL INJURY — MECHANISM AND PRESENTATION

- The kidney is the commonest urological organ injured. Blunt trauma causes 80–90%; most are grade I–III and heal conservatively.
- **Deceleration** injures the fixation points (hilum and ureteropelvic junction, UPJ), causing artery thrombosis, vein disruption, and pedicle or UPJ avulsion.
- **Penetrating** injury causes more bleeding, nephrectomy and complications; high-velocity missiles cause delayed necrosis.
- Signs: flank ecchymosis and **ipsilateral rib 9–12 fractures** (strongest association).

2 · HAEMATURIA

- Significant haematuria is **gross**, or **microhaematuria (>5 red blood cells per high-power field) with systolic blood pressure (SBP) <u>≤90 mm Hg</u>**.
- Degree does not match severity; haematuria is absent in ~⅓ of blunt renovascular injuries and **~½ of UPJ injuries**. Test the first aliquot.

KEY POINT

Haematuria triggers imaging; it does not measure injury. The worst injuries (pedicle, UPJ) may not bleed into the urine, so mechanism carries equal weight.

3 · AAST KIDNEY INJURY SCALE (2018)

The AAST grade, set on CT, predicts intervention and nephrectomy.

Grade	AAST 2018
I	Subcapsular haematoma and/or contusion, no laceration
II	Perirenal haematoma within Gerota's fascia; laceration **≤1 cm**, no extravasation
III	Laceration >1 cm, no collecting-system rupture; **any vascular injury or active bleeding contained within Gerota's**
IV	Collecting system laceration with extravasation; pelvis laceration or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's; segmental/complete infarction without active bleeding
V	**Main renal artery or vein laceration** or hilar avulsion; devascularised kidney with active bleeding; shattered kidney

- Advance one grade for bilateral or multiple injuries, up to III; grade by the highest finding. MCQ Main-vessel injury was grade IV in 1989 and is now V; segmental pseudoaneurysm or arteriovenous fistula (AVF) is now IV.

4 · INDICATIONS FOR IMAGING (AUA = EAU)

- **Penetrating** trauma with **any haematuria** needs CT, if the patient is stable.
- **Blunt** trauma needs CT with rapid deceleration, gross haematuria, or microhaematuria with SBP <90 at any time; otherwise blunt microhaematuria is **observed without imaging**.
- **Children** hold blood pressure until ~50% blood loss, so image liberally.

5 · IMAGING STUDIES

Without a delayed phase, urinary extravasation is missed.

- Gold standard: **contrast CT with arterial/nephrographic and delayed (~10 min) phases**. A medial haematoma with normal arteries suggests vein injury.
- Focused assessment with sonography for trauma (FAST) cannot separate haematoma from urine.

CT finding	Suggests
Medial haematoma	Vascular pedicle injury
Medial extravasation; no contrast in distal ureter	Renal pelvis/UPJ avulsion
Absent global enhancement (cortical rim sign)	Main renal artery occlusion
Contrast blush	Brisk bleeding; strongly predicts need for angioembolisation (AE)
≥2 of: rim distance >4 cm, medial laceration, blush	High-risk grade IV

- The **MiGUTS nomogram** predicts bleeding intervention; its largest contributor is **haematoma rim distance**.

ALGORITHM

1. **One-shot intravenous pyelogram (IVP)** (unstable, no CT): **2 mL/kg contrast → single film at 10 min**. It confirms a working contralateral kidney.

Perinephric collection	Hounsfield units (HU)	Feature
Urinoma	0–20	Contrast pooling on delayed phase
Haematoma	>30	—
Abscess	>20	Rim enhancement

6 · NONOPERATIVE MANAGEMENT

- Stable, staged **grade I–IV and selected V** injuries are observed, whatever the mechanism. Success is >90% in blunt injury, even with extravasation; early mobilisation is safe.
- **Failure triggers**: instability; transfusion of **2–3 units**; new pain or guarding; enlarging urinoma with fever. Most failures need only stent, nephrostomy or AE.
- **Penetrating** injury breaches Gerota's fascia, losing tamponade, so more fail; stabs do better than gunshots.
- **Reimaging** (AUA): CT at ~48 h for **grade IV–V deep lacerations** or deterioration, not for uncomplicated I–III.

7 · URINARY EXTRAVASATION

- Parenchymal leaks **heal in 76–90%**. **Drain early** when urine cannot pass antegrade: pelvis or UPJ avulsion, ureteric clot, UPJ obstruction.
- A symptomatic or enlarging urinoma gets a retrograde stent and catheter; if that is impossible, a nephrostomy.

> **KEY POINT**
> Parenchymal leaks heal because antegrade flow is preserved; anything stopping urine reaching the bladder converts an observable leak into one that must be drained.

8 · INTERVENTION

Needless exploration releases tamponade and costs kidneys.

Absolute indications	Relative (increasingly observed)
Instability, no or transient response	Extravasation with significant devascularisation
Expanding or pulsatile haematoma	Colon or pancreatic injury
Suspected pedicle avulsion; UPJ avulsion	Arterial thrombosis; parenchymal extravasation

- **AE** is superselective, raises salvage and can be repeated. **Failure predictors**: instability, grade V, urinary extravasation, gross haematuria.
- At laparotomy, exploration is **not mandatory** for staged grade I–III. The only absolute indication is a **pulsatile or expanding haematoma**.

ALGORITHM

1. **Vascular control first**
2. Eviscerate the small bowel; incise peritoneum over the aorta above the inferior mesenteric artery (IMA), up to the ligament of Treitz.
3. Follow the aorta to the left renal vein; loop and retract it; loop both renal arteries.
4. Reflect the colon and open Gerota's fascia; occlude the vessels if bleeding.

- **Damage control** (hypothermia, acidosis, coagulopathy): pack, close temporarily, resuscitate and return at 24–48 h.

ALGORITHM

1. Expose and gain vascular control; debride only devitalised tissue.
2. Suture-ligate vessels; close the collecting system **watertight**.
3. Approximate parenchyma over a bolster; cover with Gerota's fascia or **pedicled omentum**; drain.

- Colon or pancreatic injury no longer mandates nephrectomy.
- **Revascularise** only a solitary kidney or bilateral injury, with warm ischaemia **<120 min**; unilateral repair salvages only ~⅓.

- The **left renal vein may be ligated; the right cannot**.
- Hypertension follows thrombosis repair in ≥43%, so monitor blood pressure for a year.

9 · COMPLICATIONS AND FOLLOW-UP

- Delayed bleeding comes within 21 days and is treated by AE.
- **Page kidney** (subcapsular compression) is treated with an ACE inhibitor or receptor blocker, then drainage if refractory; post-traumatic AVF needs AE.
- EAU: no imaging for grade I–II; salvaged grade III–V get a dimercaptosuccinic acid (DMSA) scan.
- Bleeding risk peaks at **2–6 weeks**; a solitary kidney does not bar contact sport.

10 · URETERAL INJURY

Often silent; late diagnosis means fistula or kidney loss.

P3 *Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury — aetiology, risk sites, prevention, CT with delays/retrograde, damage control, repair by level (hitch 95–100%, Boari up to 12 cm, TUU, ileal ureter), follow-up*

Not in the paper answer:

- Delayed upper-ureter repair may use appendix (right) or buccal graft.

AAST ureter grade	Description
I	Contusion/haematoma without devascularisation
II	Laceration **<50%** transection
III	Laceration ≥50% transection
IV	Complete transection, **<2 cm** devascularisation
V	Avulsion, >2 cm devascularisation

- Iatrogenic injury is commoner (distal); external is mostly gunshot (proximal). **Haematuria is absent in 25–45%**.
- The **distal ureter** is the commonest iatrogenic site; prolapse surgery is highest-risk, and minimally invasive hysterectomy is riskier than open.
- **Ureteroscopy**: stricture ~3%, avulsion <1%. At a tear, stop basketing and stent.
- Preoperative stents aid **recognition**, not proven prevention. Check efflux at cystoscopy after minimally invasive hysterectomy and prolapse surgery. **Methylene blue** is a monoamine oxidase inhibitor; avoid it in pregnancy and G6PD deficiency.

Modality	Role
CT urography, **10-min delays**	Best in a stable patient; trace both ureters
Retrograde ureterography	**Most sensitive**; allows stenting
Antegrade study	When retrograde fails
One-shot IVP	Often non-diagnostic; never rely on it

- CT signs: **no contrast distal to the injury**, medial extravasation (UPJ avulsion), delayed nephrogram, urinoma.

> **KEY POINT**
> Ureteral injury is diagnosed by absent contrast beyond it, so time the study for excretion, follow both ureters, and never treat a non-diagnostic study as negative.

ALGORITHM

1. If stable at laparotomy, repair now; if unstable, do damage control and repair later.
2. If contusion, stent, or excise and repair.
3. If found late or partial, stent; if stenting fails, place a nephrostomy and repair later.
4. If ureterovaginal fistula, try a stent first.

- **Damage control**: ligate the proximal stump with long silk and place a nephrostomy once stable, or exteriorise a single-J stent.
- Principles: spare adventitia; debride to bleeding edges; make it **spatulated, tension-free, stented and watertight**; drain. **No bowel or autotransplantation acutely.**
- In trauma, reimplant **non-tunnelled**, because tunnelled reimplants stricture more.
- Psoas hitch beats distal ureteroureterostomy, because the distal stump's supply is tenuous; spare the genitofemoral and femoral nerves.

95 Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis

1 · OVERVIEW AND EPIDEMIOLOGY

- **Removal is surgical, but recurrence is metabolic**. Prophylaxis: see Ch 99.
- Lifetime prevalence is **1–15%**; in the United States ≈ **10%** and rising.
- **Sex**: males outnumber females 2–3 : 1, but the gap is narrowing. **Oestrogen is protective**, hence a second female peak at menopause.
- **Race**: highest in White, lowest in Asian people. **Age**: peak in the 4th–6th decades; paediatric stones are rising.
- **Climate and occupation**: heat lowers urine volume and pH, sunlight causes vitamin D-mediated hypercalciuria, and sedentary work causes hypercalciuria.
- **Obesity**, **metabolic syndrome** and **diabetes** raise risk. **Insulin resistance** unifies these: it lowers urine pH, causing uric acid stones, and hyperinsulinaemia causes hypercalciuria.
- Hypertension, cardiovascular disease and chronic kidney disease (CKD) are associated. Fluid intake is inversely related (proven by randomised controlled trial, RCT).

2 · PHYSICOCHEMISTRY AND STONE FORMATION

Prevention lowers supersaturation or aids inhibition.

P2 *Stone formation (etiopathogenesis) and medical management of urolithiasis — saturation states, heterogeneous nucleation, steps, Randall plaque vs ductal plugs, inhibitor list, matrix 2.5%*

Not in the paper answer:

- Free-particle theory (crystals trapped as plugs) vs fixed-particle theory (anchored on plaque).
- Stones form by **supersaturation, then nucleation, growth and aggregation, then retention**. The concentration product (CP) multiplies ion concentrations; the solubility product (Ksp) is CP at saturation; the formation product (Kf) is CP at spontaneous crystallisation.

State	CP	What happens
Undersaturated	< Ksp	No crystals; stones may dissolve
Metastable	Ksp–Kf (most urinary salts)	No homogeneous (spontaneous) nucleation; growth and heterogeneous nucleation possible; **inhibitors effective, so this is the zone of therapy**
Unstable	> Kf	Spontaneous nucleation; inhibitors ineffective

- The **relative saturation ratio (RSR)** is CP ÷ Ksp. Below 1, stones dissolve; above 1, they grow.

- Changing pH lowers the saturation of **phosphate, uric acid and cystine, but not oxalate**. **Calcium and oxalate contribute equally** to calcium oxalate saturation.
- Urine holds calcium oxalate at ≈ 4× its solubility in water; with inhibitors it precipitates only at **7–11×**.
- Nephron transit is too short for one crystal to block a tubule, so aggregation occludes.
- **Randall plaque** (MCQ 1937) is **calcium apatite**. It starts in the **thin-limb basement membrane**, spreads through the interstitium and erodes the papilla; calcium oxalate then grows on it.

Stone former	Randall plaque	Ductal plugs
Idiopathic calcium oxalate	Present, as the anchor	Uncommon, mild
Intestinal bypass	Absent	Apatite plugs in inner medullary collecting ducts; injury, fibrosis
Brushite	Present	Apatite plugs; interstitial fibrosis
Distal renal tubular acidosis (RTA), hyperparathyroidism, cystinuria	—	Present

KEY POINT

Plaque vs plug: a plaque is an idiopathic interstitial apatite anchor; a plug is intratubular obstruction with injury, caused by systemic disease. **Ductal plugs should prompt a search for a secondary cause.** Infection or shock-wave lithotripsy may convert a calcium oxalate former into a brushite former.

3 · INHIBITORS AND PROMOTERS

- Inhibitors raise the supersaturation needed for nucleation. **Uric acid and cystine have no inhibitor**; only pH and concentration matter. Promoters: low volume and citrate; high calcium, oxalate, phosphate, uric acid.

Inhibitor	Action	Key feature
Citrate	Complexes calcium; inhibits precipitation, agglomeration and urate-induced nucleation	Most important small-molecule inhibitor
Magnesium	Complexes oxalate	Synergy with citrate, phytate
Glycosaminoglycans	Bind crystal surface	Lower in stone formers
Nephrocalcin	Inhibits calcium oxalate monohydrate nucleation, growth, aggregation	Formers excrete isoforms lacking γ-carboxyglutamic acid
Tamm-Horsfall protein	Inhibits aggregation, not growth	Most abundant urinary protein, made in the thick ascending limb (TAL); **a promoter in acid urine**
Osteopontin	Inhibits nucleation, growth, aggregation, epithelial binding	Inducible

- **Matrix** is ≈ **2.5%** of stone weight.

4 · MINERAL METABOLISM

- **Calcium**: 30–40% is absorbed in the small bowel, transcellularly (calcitriol-dependent) at low intake and paracellularly at high intake.
- **Oxalate-rich food lowers calcium absorption, and luminal calcium lowers oxalate absorption**.
- **Calcitriol**, made in the proximal tubule under **parathyroid hormone (PTH) and hypophosphataemia**, is the most potent stimulator of gut calcium absorption.

- **PTH** causes bone resorption, raises renal calcium reabsorption and calcitriol, and lowers phosphate reabsorption; it has **no direct gut effect**.

Segment	Filtered calcium reabsorbed	Route
Proximal tubule	**60–65%**	Paracellular; claudin-2
TAL	25–30%	Paracellular, lumen-positive; claudin-16/19; the calcium-sensing receptor (CaSR) acts through claudin-14 to block it
Distal convoluted tubule (DCT)	8–10% (fine control)	Transcellular: entry by TRPV5 channel, then carried by calbindin-D28k

- **Oxalate**: 6–14% is absorbed, unregulated, and excreted almost entirely in urine; half is dietary, half hepatic. **Ascorbic acid** is the largest endogenous source; Oxalobacter formigenes degrades oxalate.

5 · CLASSIFICATION

Stone	Frequency
Calcium oxalate (pure or with apatite)	**60%**
Hydroxyapatite	20%
Brushite	2%
Uric acid	≈ 7% (8–10% US, rising)
Struvite	≈ 7% (falling)
Cystine	1–3% (up to 10% children)
Triamterene, silica, 2,8-dihydroxyadenine (2,8-DHA)	< 1% each

Condition	Defect	% of formers
Absorptive hypercalciuria (incl. phosphate leak)	Increased gut calcium absorption	20–40
Renal hypercalciuria	Renal calcium leak	5–8
Resorptive hypercalciuria	Primary hyperparathyroidism (PHPT)	3–5
Hyperuricosuric calcium	Purine excess	10–40
Hypocitraturic calcium	Idiopathic, diarrhoea, distal RTA, thiazide	10–50
Hyperoxaluric calcium	Primary, enteric, dietary	2–15
Hypomagnesiuric calcium	Reduced gut magnesium absorption	5–10
Idiopathic low urine pH	Low pH	15–30
Low urine volume	Inadequate intake	10–50
Infection; cystinuria	Urease; cystine transport	1–5; < 1
No abnormality	—	< 3

> **KEY POINT**
> Classify by the **urinary abnormality, not the stone**: composition starts the work-up. Fewer than 3% have no abnormality. Work-up: see Ch 99.

6 · HYPERCALCIURIA

Fasting urine and PTH reveal the source.

- It is the commonest abnormality (35–65%).
- **Definition**: > **200 mg/day** after 1 week on a 400 mg calcium, 100 mEq sodium diet; or > 4 mg/kg/day.

- The **Pak classification** (gut, kidney, bone) is an exam framework, not a therapy guide.

Feature	Absorptive	Renal	Resorptive
Defect	Increased gut absorption	Renal leak	PHPT
Serum calcium	Normal	Normal	**High**
PTH	**Suppressed**/normal	**High** (secondary)	High
Fasting urinary calcium	**Normal** (< 0.11 mg/dL glomerular filtrate)	**High**	High
Serum phosphate	Normal (low in phosphate leak)	Normal	Low
Sequence	Hyperabsorption suppresses PTH and raises the filtered load	Renal loss raises PTH, which raises calcitriol	PTH drives bone resorption and calcitriol, causing hypercalcaemia

- Absorptive **type I** persists on calcium restriction; **type II** normalises.
- **CYP24A1** loss stops calcitriol inactivation, causing hypercalciuria and nephrocalcinosis. In **renal phosphate leak**, hypophosphataemia raises calcitriol and gut calcium absorption.

Disorder	Gene	Features
Dent disease (X-linked)	ClC-5	Hypercalciuria, **low-molecular-weight proteinuria**, nephrocalcinosis, renal failure
Familial hypomagnesaemia with hypercalciuria and nephrocalcinosis (FHHNC)	Claudin-16, -19	TAL magnesium/calcium loss
Bartter	NKCC2, ROMK, ClC-Kb, barttin; type V activating CaSR	Salt wasting, hypokalaemic alkalosis, hypercalciuria

KEY POINT

Fasting urine and PTH separate gut from kidney: in renal leak, fasting calcium and PTH are high; in absorptive hypercalciuria, fasting calcium is normal and PTH suppressed.

- **PHPT** causes ≈ 5% of stones, and stones occur in < 5% of PHPT. Suspect it if serum calcium is > **10.1 mg/dL**; repeat with intact PTH, and use ionised calcium for normocalcaemic PHPT. A **thiazide challenge** unmasks it.

Cause	Mechanism	PTH	Clue
PHPT	Autonomous PTH	High/inappropriately normal	Commonest in outpatients
Sarcoidosis	Macrophage 1α-hydroxylase	Suppressed	Corrects with **corticosteroids**
Malignancy	PTH-related protein (PTHrP), osteolysis (RANKL)	Suppressed	Commonest in inpatients
Glucocorticoid excess	Increased bone resorption	Raised	Cushing syndrome

7 · HYPEROXALURIA

- **Definition**: > **40 mg/day**. Types: primary, enteric, dietary and idiopathic.

P2 *Primary Hyperoxaluria — PH1–3 enzymes, markers, course; oxalate ≥100 mg/day, oxalosis; pyridoxine (Gly170Arg); lumasiran; liver–kidney transplant in PH1*

ALGORITHM

1. **Enteric hyperoxaluria**
2. Fat malabsorption leaves fatty acids in the gut lumen.
3. Fatty acids saponify calcium and magnesium, so less calcium binds oxalate and more oxalate is free.
4. Free oxalate undergoes **colonic hyperabsorption**, helped by bile salts.

- **Enteric** is the commonest acquired form: small-bowel resection or Crohn disease, jejunoileal bypass and malabsorptive bariatric surgery (not band or sleeve).
- It **needs an intact colon**; with an ileostomy, uric acid or ammonium urate stones form instead. Diarrhoea also causes hypocitraturia, hypomagnesiuria and low urine pH.
- **Dietary**: nuts, chocolate, tea, spinach, rhubarb and vitamin C. **Severe calcium restriction is counter-productive**, as less calcium binds gut oxalate.

8 · HYPERURICOSURIA, HYPOCITRATURIA, HYPOMAGNESIURIA

- **Hyperuricosuria** is **> 600 mg/day**. Urate promotes calcium oxalate stones by **adsorbing inhibitors, "salting out" and epitaxy**; allopurinol cut recurrence (trial below).
- **Hypocitraturia** is **< 320 mg/day**; **acid–base status is the main determinant**. Causes: distal RTA, chronic diarrhoea, animal protein, thiazides (through hypokalaemia), ACE inhibitors.

KEY POINT

Citrate excretion reads the proximal-tubular **intracellular pH**: acidosis, hypokalaemia or an acid diet make the tubule reclaim citrate. Hence potassium citrate is the treatment (see Ch 99).

- Hypomagnesiuria: **two RCTs of magnesium showed no benefit**. **Low urine pH** (< 5.5) causes uric acid and calcium oxalate stones.

9 · RENAL TUBULAR ACIDOSIS

P2 *Renal tubular acidosis — type 1/2/4 comparison (urine pH, K^+, citrate, stones up to 70%), incomplete dRTA, ammonium chloride load, topiramate/acetazolamide*

Not in the paper answer:

- Genetics: SLC4A1 (dominant); ATP6V1B1/ATP6V0A4 (recessive, with **sensorineural hearing loss**).

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4
Defect	Distal H^+ secretion	Proximal HCO_3^- reabsorption	Aldosterone deficiency/resistance
Serum K^+	Low	Low	**High**
Urine pH	**Always > 6.0**	< 5.5 at steady state	Low
Urinary citrate	**Profoundly low**	Near normal	—
Stones	**Up to 70%**: calcium phosphate, nephrocalcinosis	Uncommon	Uncommon
Associations	Sjögren, lupus, obstruction; genetic	Fanconi	Diabetic nephropathy

- **Incomplete distal RTA**: electrolytes are normal, but urine pH cannot fall below 5.5 after an ammonium chloride load.
- In distal RTA, **profound hypocitraturia** is probably the most important stone factor.

KEY POINT

Stone risk in RTA follows citrate and pH: only distal RTA combines fixed alkaline urine with profound hypocitraturia and hypercalciuria, the recipe for **calcium phosphate** stones.

10 · URIC ACID STONES

P2 *Uric acid calculi: etiopathogenesis, medical management and allopurinol — no uricase, pKa 5.35, three determinants (low pH dominant), insulin resistance/ammoniagenesis, idiopathic vs hyperuricosuric calcium*

Not in the paper answer:

- Uric acid forms about one-third of stones in type 2 diabetes.

Determinant	Causes	Importance
Low urine pH	Insulin resistance/obesity, animal protein, diarrhoea, gout	**Single most important**
Low volume	Diarrhoea, heat	Concentrates uric acid
Hyperuricosuria	Purine diet, gout, myeloproliferative disease, uricosurics	Least often the driver

- Humans lack hepatic **uricase**, so uric acid is not converted to soluble allantoin.
- The pKa of uric acid is **5.35**. Undissociated uric acid dissolves only to ≈ 100 mg/L, but at pH 6.5 > 1200 mg/L stays soluble.
- Stones are radiolucent on plain film but visible on non-contrast CT.

ALGORITHM

1. **Why the urine is acid in insulin resistance**
2. Insulin resistance and lipotoxicity lower proximal ammoniagenesis and sodium–hydrogen exchanger 3 (NHE3) activity.
3. H^+ is excreted as titratable acid instead of ammonium.
4. Urine pH stays **persistently low**, uric acid stays undissociated, and stones form.

KEY POINT

Uric acid stones are a disease of **urine pH, not uric acid**. No inhibitor exists, so raising pH above the pKa removes the driver; alkalinisation can dissolve stones (see Ch 99).

11 · CYSTINE AND STRUVITE STONES

Both recur aggressively without lifelong prevention or complete clearance.

P1 **P4** *Cystinuria; Struvite stones — SLC3A1/SLC7A9 types, solubility by pH, hexagonal crystals, CKD; urease cascade, biofilm, Proteus/Ureaplasma, discordant cultures*

Not in the paper answer:

- **Cystinuria genes**: SLC3A1 (type A), SLC7A9 (type B), or both (type AB). It is autosomal recessive, and with **no inhibitor** cystine is soluble only at alkaline pH (≈ 1000 mg/L at pH 9).
- **Struvite**: **Proteus mirabilis** is commonest; **Ureaplasma** is easily missed. **Less than half of staghorns are infection stones**, and bacteria within the stone demand complete clearance.

ALGORITHM

1. **Urease cascade**
2. Urease splits urea into ammonia (NH_3) and CO_2.
3. Ammonia forms NH_4^+ and OH^-, raising urine pH to 7.2–8.0.
4. With magnesium, phosphate forms **struvite**; with calcium, carbonate apatite.

12 · MISCELLANEOUS AND DRUG-RELATED STONES

	Xanthine	2,8-DHA
Enzyme deficient	Xanthine dehydrogenase (XDH)	Adenine phosphoribosyltransferase (APRT)
Radiology	Both radiolucent, so mistaken for uric acid	
Treatment	—	Allopurinol or febuxostat (more effective)

- **Ammonium acid urate** stones are radiolucent and follow diarrhoea, ileostomy, **laxative abuse** and infection.

Route	Drug	Mechanism / stone
Direct	**Indinavir**	Insoluble at physiological pH; stones 4–13%; radiolucent even on CT
Direct	Triamterene; guaifenesin/ephedrine; silicates	Drug stones
Indirect	Acetazolamide, **topiramate**, zonisamide	Mimic distal RTA, causing calcium phosphate stones
Indirect	Loop diuretics	Hypercalciuria (up to 64% of low-birth-weight infants on furosemide)
Indirect	Steroids, vitamin D, thiazides, laxatives, cytotoxics	Hypercalciuria, hypocitraturia, urate

13 · ANATOMIC PREDISPOSITION AND PREGNANCY

Anomaly	Stones	Metabolic findings
Ureteropelvic junction obstruction (UPJO)	≈ 20%; children ~70× risk	Common (hypercalciuria first); recurrence after repair, often contralateral
Horseshoe kidney	≈ 36% adults	All tested abnormal; **hypocitraturia**; calcium stones ≈ 89%
Calyceal diverticulum	Up to 40%	Abnormal in half; UTI up to 40%
Medullary sponge kidney	Frequent, with nephrocalcinosis	Hypercalciuria, hypocitraturia; distal RTA on acid loading

> **KEY POINT**
> Anatomy sets the stage but metabolism writes the stone: anatomical stone formers still need **metabolic evaluation**, and pyeloplasty alone does not prevent recurrence (see Ch 99).

- **Pregnancy**: stones occur in the 2nd–3rd trimester. Physiological hydronephrosis affects up to 90%, **right more than left**.
- Raised GFR and placental calcitriol promote stones, while raised citrate and magnesium inhibit them; the **net risk is unchanged**, but stones skew to **calcium phosphate**.

14 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Borghi water trial	RCT, high fluid vs no advice, first-time idiopathic calcium formers	Recurrence 12% vs 27%	Urine volume is the one proven universal modifier (see Ch 99)
Ettinger allopurinol trial	RCT, allopurinol vs placebo, hyperuricosuric normocalciuric calcium oxalate formers	> 2-fold fewer recurrences	Allopurinol for hyperuricosuric calcium oxalate stones (see Ch 99)

96 Acute Care of Urinary Lithiasis

1 · PRESENTATION AND DIAGNOSIS

Fever with obstruction is an emergency.

- **Renal colic** is flank-to-groin pain; a UVJ stone causes penile-tip pain and frequency.
- Haematuria occurs in ~**90%**, so it is absent in >10%.
- **Fever means an infected obstructed system**, needing urgent intervention.

- Send bloods, urinalysis and a **pregnancy test** in women of reproductive age. Urine >50 WBC/hpf or nitrites suggest infection.

Modality	Role	Limitation
Non-contrast CT (NCCT)	Gold standard (sens 97%, spec 95%); shows alternative diagnoses	8–16 mSv; misses matrix and indinavir stones
Low-dose NCCT (≤**3 mSv**)	Equal for stones ≥3 mm	Weak <3 mm and in obesity
Ultrasound (US)	First line in pregnancy, children, known formers	Up to 40% still need CT
Plain X-ray (KUB)	**Avoid** for acute diagnosis	Lowest accuracy
MRI	Second line in pregnancy and children	Poor stone detection

KEY POINT

Unknown diagnosis: NCCT answers everything at once. Known stone: US showing hydronephrosis suffices.

2 · PAIN MANAGEMENT

- **NSAIDs** block prostaglandin release and match opioids with less vomiting.

Agent	Place	Note
Ketorolac 30 mg	**First line** (EAU, NICE)	Avoid in renal failure, GI bleed, pregnancy
IV paracetamol	Second line	NSAID barred or insufficient
Opioids	Intractable pain	**Continue the NSAID**: morphine plus ketorolac beats either
Antispasmodics; forced fluids	**Ineffective**	Fluids only for vomiting

3 · MEDICAL EXPULSIVE THERAPY (MET)

P2 *Medical expulsive therapy (MET) in urinary calculi — α1 mechanism, tamsulosin/silodosin, passage-by-width table, SUSPEND/STONE, benefit in distal 5–10 mm, intervene after 4–6 weeks*

Not in the paper answer:

- Measure **transverse width**, not length; a 6 mm wide stone passes in ≈ 33%.
- Trial of passage, with or without MET, lasts **4–6 weeks**.

4 · URGENT INTERVENTION

- **AUA indications**: obstruction with infection, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration, refractory pain or vomiting.
- **Stent and nephrostomy (PCN)** are equal for sepsis; stents cause dysuria and haematuria, PCN a longer stay.
- **Stent failure** is predicted by P-CUT (ureteric diameter at the stone minus stone width) ≥**7.5 mm**, stone ≥9.5 mm and multiple stones; prefer PCN then.
- Acute SWL (proximal) or URS (distal <10 mm) is done **only without infection**.

ALGORITHM

1. Give analgesia; get NCCT.
2. If infection or an AUA criterion, then drain only (stent/PCN); treat the stone later.
3. If not and stone <10 mm, then trial with or without MET for 4–6 weeks.
4. If it fails, then URS or SWL.

5 · SPECIAL POPULATIONS

P2 *Stones (urolithiasis) in pregnancy — US, then MRI, then low-dose CT; conservative first, stent/PCN encrustation, URS any trimester*

Not in the paper answer:

- Resistive index difference ≥**0.06** means obstruction.
- About 30% need intervention; change stent/PCN every **4–6 weeks**. URS is safe in any trimester (ACOG 2019).

Population	Pitfall	Key point
Elderly	Atypical pain; first stone after 60 unusual	Always CT to exclude aortic/cardiac mimics; drain early
Transplant kidney	Denervated, so **painless**	Fever or renal failure; endoscopic, else percutaneous

6 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SUSPEND	RCT, tamsulosin vs placebo, any ureteral stone, n = 1136	Intervention at 4 weeks 19% vs 20%	Unstratified; does not abolish MET for distal 5–10 mm

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P2 *Management of upper ureteric stones — passage rates, EAU choice by size, SWL vs URS table, SWL failure factors, flexible URS, antegrade and laparoscopic routes, complication rates*

Not in the paper answer:

- SWL **within 48 h** of an acute obstructing stone beats delayed SWL.
- Little is gained beyond 2 SWL sessions; **forced diuresis is useless**.

1 · BACKGROUND AND INDICATIONS

- Non-contrast CT (NCCT) is the gold standard; ultrasound misses mid-ureteric stones. US with a plain film (KUB) suits follow-up.
- Medical expulsive therapy is controversial after SUSPEND; intervene after **4–6 weeks**.
- The ladder is SWL, then ureteroscopy (URS), then percutaneous nephrolithotomy (PCNL) with antegrade URS.

Tier	Trigger
Elective	Failed trial of passage
Urgent	Unrelenting pain; obstruction-related renal dysfunction
Extremely urgent	Obstructive pyelonephritis or sepsis: **drain only**, never manipulate the stone

- Work-up: creatinine and **urine culture**; exclude coagulopathy and untreated UTI.

Modality	Antiplatelet/anticoagulant	Bridging
PCNL	Stop if feasible	Short-acting heparin if high risk
SWL	Stop if feasible	Not recommended; choose another modality
Laser URS	**Need not stop**	None; more haematuria and stent symptoms

KEY POINT

Only URS tolerates anticoagulation: SWL and PCNL injure parenchyma where bleeding is unseen; laser URS works inside the lumen under vision.

2 · CHOOSING TREATMENT

- Upper ureter ≤2 cm and distal ureter: stone-free rate (SFR) is similar, but SWL needs more sessions.
- **Impaction** (thick wall around the stone on NCCT) defeats SWL, so treat endoscopically.
- A large hard proximal stone needs PCNL with antegrade URS; if unfit, staged URS.

Modality	Size guidance	
SWL	Historical limit 2 cm; SFR falls above	**10–15 mm**
URS	No definite limit	
PCNL with or without antegrade URS	>15–20 mm, large hard proximal	

KEY POINT

Constant trade-off: SWL means fewer complications but more sessions; URS means one session with slightly more morbidity.

3 · URETEROSCOPES

	Semirigid	Fibreoptic flexible	Digital flexible	Single-use digital
Image	Stable unless bent	Degrades as fibres break	No degradation	New each case
Calibre	Tapered	Smallest tip	~8.5 Fr	7.7 Fr
Cost	Cheap	Repair ~every **15** cases		Low volume

- Semirigid scopes suit the distal and mid ureter; flexible scopes suit proximal stones.
- Digital scopes are **not more durable**; keep a fibreoptic scope for a narrow ureter.

4 · ACCESS

Most injuries happen at access, so be safe before fast.

P3 *Accessories for RIRS; Ureteric stents and ureteroscopy with or without stenting — guidewire types, 12/14 Fr sheath, safety wire (Segura), pre-stent 2–4 weeks, stent indications, Denstedt stentless trial*

Not in the paper answer:

- **α-Blockers** ~1 week pre-URS give easier access, higher SFR and less stent pain.
- Reserve the sheath for stones **above the iliac vessels**.
- For a difficult ureter, place a stent and return after **2–4 weeks**.

ALGORITHM

1. If the wire will not pass, place an open-ended catheter 1–2 mm short of the stone.
2. Do a retrograde pyelogram.
3. Use contrast hydrodissection with a hydrophilic wire.
4. Advance the catheter past the stone, then exchange for a stiff wire.

- If the catheter abuts the stone, the wire exits at 90° and causes **perforation**. If the ureter is inaccessible, **stent and stage**.
- If the scope will not enter the orifice, use **railroading** between two wires, a dual-lumen catheter or scope rotation.

Dilation method	Best for
Indwelling stent	Whole ureter narrow; gentlest, but commits to a second procedure
Serial/coaxial dilators	Long narrow segments
Balloon	Focal stenosis; inflate until the waist resolves, hold 1–2 min
Semirigid scope	Distal ureter

- **Pre-stenting is not needed** for ureteral stones; it helps only renal stones.
- The access **sheath** helps with large burden, dusting and high power, lowering intrarenal pressure (IRP) and heat.

5 · LITHOTRIPSY

- **Basketing**: keep the stone **5–10 mm from the tip**, **never basket blind**, and when in doubt **fragment first**.

Source	Key points
Electrohydraulic	High ureteral trauma; abandoned
Pneumatic	~12 Hz; **rigid/semirigid only**; best below brim; retropulsion
Holmium laser (Ho:YAG)	Short pulse fragments; long or modulated pulse lowers retropulsion; dusting is low energy, high frequency
Thulium fibre laser (TFL)	Low peak power and long pulse give fine dust; minimal retropulsion; >40 Hz misses

P4 *Thalium laser in urology — TFL physics, settings (0.6 J × 6 Hz; dust 0.2 J × 40 Hz), "paint, do not drill", RCTs vs holmium*

Not in the paper answer:

- Impacted stone: target the centre first, so the sides fall away from the wall.

Setting	Ho:YAG	TFL
Fragmentation	0.6–1.0 J × 6–10 Hz	0.6 J × 6 Hz short pulse (3.6 W)
Dusting	0.2–0.3 J × 40–80 Hz	0.2 J × 40 Hz (8 W)

- **Settings, not only the laser, decide outcome**.
- **Thermal** damage begins above **43 °C**, and **heat tracks power, not laser type**.
- Cap at **≤10 W in ureter** and ≤25 W in kidney, with continuous irrigation, because stopping it causes thermal stricture.

6 · AFTER URS

Most need neither long antibiotics nor a stent.

- Give a **single perioperative intravenous antibiotic dose**; extend only after urosepsis this episode.
- Stents cause symptoms in ~4/5; stentless uncomplicated URS is safe.
- Stent for perforation, stricture, tight ureter, infection or a second stage.

7 · SWL FOR URETERAL CALCULI

- SWL is an option at all levels.

Absolute contraindications	Relative
Pregnancy; uncontrolled bleeding disorder; anticoagulation that cannot stop (for treatment and 48 h); distal obstruction; untreated UTI; uncontrolled hypertension	Skeletal deformity or obesity preventing focusing; adjacent aneurysm

- **Predictors**: size matters most, and URS is better ≥10 mm. Impaction (CT "ring" sign) is possibly the strongest.
- Cystine, brushite and calcium oxalate monohydrate resist SWL. Density >1000 HU and skin-to-stone distance >10 cm are proven for renal stones only.
- **No routine pre-stenting**. Hold aspirin, NSAIDs and anticoagulants for treatment and 48 h. Give antibiotics only with stents, struvite or recent UTI.

- Position the upper ureter supine, the mid-ureter **prone** and the distal ureter supine; **60–90 shocks/min** beats 120. α-Blockers raise SFR.

8 · SPECIAL SITUATIONS

- **Pregnancy**: treat conservatively first; **SWL is contraindicated**; URS is safe with shielded fluoroscopy. Stents encrust, so change them **every 4 weeks**.
- **Diversion or reimplant**: use SWL (prone) or **antegrade flexible URS**. Transplant: both routes are safe.

9 · COMPLICATIONS

- **Perforation** (0–4%): stop, stent for ~4 weeks and drain any urinoma.
- **Avulsion** (0.06–0.5%) follows forceful extraction, mostly in the **proximal third**; place a **nephrostomy and repair later**.
- **Stricture** occurs in 3–6% and is silent, so image every patient.
- **Sepsis** (~5%) is linked to positive culture, pre-stenting, diabetes and long surgery.

> **KEY POINT**
> Nearly every serious URS complication is force without vision: blind basketing, forced sheaths, persistence in a tight ureter. Remedy: see, fragment, and when in doubt stop, stent and stage.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Stentless URS	RCT, stent vs none after uncomplicated URS	Safe without stent	Stent only for named indications
TFL vs Ho:YAG	RCT, low TFL settings	TFL faster; equal ureteral SFR	Low settings optimal for TFL

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1 · CHOOSING THE APPROACH

- The framework is the 2016 AUA/Endourological Society guideline, built on index patients and shared decision making.

Domain	Factors
Stone	Size, number, location, composition
Renal anatomy	Obstruction or stasis, hydronephrosis, ureteropelvic junction obstruction (UPJO), calyceal diverticulum, horseshoe or ectopic kidney, lower pole
Patient	Infection, obesity, habitus deformity, coagulopathy, age, hypertension, renal failure or transplant, solitary kidney, diversion, pregnancy

> **KEY POINT**
> Only the stone domain is changed by choosing a modality; anatomy and patient factors decide which modality is available. Obesity, coagulopathy and pregnancy each exclude a modality, not surgery.

Asymptomatic renal stones

- The natural history is not benign: **~50%** have a stone event within **5 years**. **Size predicts intervention**; risk rises above 5 mm and above 10 mm.

- With **recurrent UTI and a non-obstructing stone**, remove it whatever its composition, and **culture the stone**.
- Treat residual fragments **>4 mm** after ureteroscopy (URS).
- **Treat small asymptomatic contralateral stones** during surgery on the symptomatic side.

Preoperative evaluation

- AUA: non-contrast CT (NCCT) **should** precede percutaneous nephrolithotomy (PCNL), and **may** guide the choice of shock wave lithotripsy (SWL) versus URS.
- SWL works best when density is **<900 HU** and skin-to-stone distance (SSD) is **<10 cm**.
- Everyone needs urine culture within 1 month.

Modality selection

Scenario (AUA)	Offer	Caveat
<2 cm, non-lower pole, symptomatic	SWL or URS	SWL has fewer complications; URS has higher stone-free rate (SFR)
Total burden >2 cm	**PCNL**; staged URS an option	SWL not first-line
Lower pole >1 cm	PCNL (highest SFR) or URS	SWL not first-line, because fragments clear poorly
Lower pole <1 cm	SWL or URS	—
Staghorn	PCNL	—
Cannot stop antiplatelet/anticoagulant	**URS**	Avoid SWL and PCNL

- EAU: for >20 mm use PCNL, then URS or SWL; for **10–20 mm, SWL and endourology are in equipoise**; for <10 mm use SWL or URS, then PCNL.

Stone morphometry scores

P1 *Stone complexity scores — Guy's, STONE, CROES, S-ReSC predict SFR, not complications; Triple D predicts SWL*

2 · SHOCK WAVE LITHOTRIPSY

P2 *ESWL: mechanism, indications, factors affecting outcome, complications and current status — peak **30–100 MPa**; sources; mechanisms (dynamic squeezing strongest); **60–90 shocks/min**, ramping, coupling; haematoma, steinstrasse (no routine stent); refuted effects; burst wave*

Not in the paper answer:

- **Absolute contraindications** are pregnancy, uncorrected coagulopathy or antithrombotics, untreated UTI, severe untreated hypertension, aortic or renal artery aneurysm (aorta **>4 cm**), obstruction distal to the stone, and inability to target the stone.
- No prophylactic antibiotic is needed if urinalysis is negative and there are no risk factors (AUA, EAU).
- The electrohydraulic HM3 (Dornier, 1980) remains the efficacy benchmark MCQ.

3 · URETEROSCOPY (RETROGRADE INTRARENAL SURGERY, RIRS)

URS is the workhorse for small stones and antithrombotic patients.

- Pre-treat bacteriuria and give a single antibiotic dose.
- **Antithrombotics**: URS is the modality for those who must continue them; antiplatelets increase admissions.

Instruments and principles

- **Safety wire**: AUA and EAU recommend it, although a multicentre study found no outcome difference. Never force a wire past an impacted stone.
- The **ureteral access sheath (UAS)** lowers intrarenal pressure but gives **no clear fall in infection**. It injures the ureter in nearly half, less if **pre-stented**.
- **Anti-retropulsion** devices reduce stone migration.
- AUA allows **stent omission** only when all four apply: normal anatomy, normal renal function, no ureteral injury and no planned second stage.

Success

"Clinically insignificant fragments" cause reintervention in up to a fifth, so report zero-fragment rates.

Complications

Complication	Rate	Risk factors / management
Sepsis	~5%	Pre-stent, positive culture, ischaemic heart disease, age, long surgery, diabetes; treat the urine first
Subcapsular/perirenal haematoma	<1%	Hydronephrosis, thin cortex, long surgery, hypertension, UTI
Stricture	~3%	UAS injury, impacted stone; ultrasound surveillance (rate underestimated)
Perforation	Few %	Stop and stent
Avulsion	<1%	Proximal ureter; immediate nephrostomy, delayed reconstruction

4 · PERCUTANEOUS NEPHROLITHOTOMY

PCNL clears the most stone; safe access is the key skill.

- Landmarks MCQ: Fernström and Johansson performed the first percutaneous stone removal in 1976.
- For 1–2 cm stones PCNL gives higher SFR than SWL or URS. It is the treatment of choice for **staghorn** and most stones >2 cm.

PCNL first-line	Applications
Burden	Stones >2 cm, staghorn, matrix stones
Lower pole	>1 cm; lower pole diverticulum or infundibular stenosis
Poor retrograde route	Urinary diversion, inaccessible lower tract, prior ureteral stricture, injury or reconstruction
Coexisting pathology	UPJO; infected stones in non-ambulatory patients

Preoperative

- Stop anticoagulants with the prescriber (bridge if high risk); **low-dose aspirin is safe**.
- **NCCT is the gold standard**, showing calyceal anatomy, SSD, HU and the surrounding organs.
- For a poorly functioning kidney, get a renogram; below 10–20% function, nephrectomy may suit.
- Give single-dose, culture-directed prophylaxis. **Tranexamic acid** reduces transfusion without more thromboembolism.

Positioning

P1 P2 *Prone vs Supine PCNL / patient positions in PCNL — prone: upper pole access, ocular/brachial plexus risk; supine: airway, gravity drainage, ECIRS*

Access: the most important step

- The **ideal** target is a posterior calyx with a short, wide infundibulum.

- **Supracostal (above the 12th rib)** upper pole access reaches most of the system but raises chest complications. Lower pole access risks colonic injury.

Step	Eye-of-the-needle (bulls-eye)	Triangulation
Align	C-arm rotated 15–30° toward surgeon; needle seen end-on	Mediolateral in the oblique view, cephalocaudal in the anteroposterior (AP) view
Depth	Rotate back past AP	Short advances, alternating views

- Confirm entry by urine return, not contrast.
- **Ultrasound access** avoids radiation and shows adjacent organs.
- In **endoscopic combined intrarenal surgery (ECIRS)**, a ureteroscope watches the puncture, giving fewer tracts and higher clearance.

Dilatation and tract size

- Dilate with Alken metal, Amplatz serial or **balloon** dilators (one pass, 20–30 atm), keeping a safety wire.
- Multiple tracts raise bleeding and reduce function, especially in chronic kidney disease.

P4 *Miniperc: miniaturised PCNL, its nomenclature and newer PCNL techniques — standard 24–34, mini 15–22, ultra-mini 11–14, micro <10 Fr; mini: equal SFR, less bleeding*

Completion and exit

- Finish with **flexible nephroscopy or URS**.

ALGORITHM

1. Confirm the collecting system and ureter are clear.
2. Withdraw the sheath under vision, with the safety wire out last.
3. Compress the tract.
4. If bleeding persists, re-inflate the balloon or place a **Kaye tamponade catheter**, and alert the anaesthetist.
5. If it still persists, arrange **selective angioembolisation**.

- Drain with a nephrostomy (≤10 Fr means less pain) or a double-J stent, or go tubeless.

Complications

- Cochrane (42 trials): **PCNL equals RIRS for major complications**, with fewer secondary procedures.
- Complications include haemorrhage (delayed bleeding from arteriovenous fistula or pseudoaneurysm needs angioembolisation), visceral injury, hydrothorax or pneumothorax, and sepsis.

5 · SPECIAL ANATOMY

Anatomy	Key adaptation
Horseshoe (stones in ~1 in 5)	SWL: long SSD. PCNL: access is medial and caudal; the upper pole posterior calyx is usually subcostal; more colonic injury; UPJO is frequent
Calyceal diverticulum/infundibular stenosis	Treat only if pain, UTI or haematuria. URS for upper or interpolar (laser-incise or dilate the ostium); lower pole: PCNL. Fulguration controversial
Pelvic kidney	SWL or URS <2 cm; PCNL with ultrasound, with or without CT or laparoscopic guidance
Urinary diversion	Antegrade wire access most reliable
Transplant	Always remove stones. The denervated graft presents with fever, oliguria and haematuria, not colic. Ultrasound-guided PCNL
Spinal deformity	SWL poor; URS good; PCNL equal clearance but more infection

- **Laparoscopic or robotic** surgery is only for when SWL, URS and PCNL fail or are expected to.

6 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Lower Pole I	Randomised controlled trial (RCT), PCNL vs SWL, lower pole >1 cm	SFR 91% vs 21%	SWL not first-line for lower pole >1 cm
Contralateral stone RCT (2022)	Remove vs observe small asymptomatic stones during surgery	Relapse 16% vs 63%	Treat contralateral stones
Cochrane PCNL vs RIRS (2023)	42 trials	No difference in major complications; fewer secondary procedures	PCNL no longer "more morbid"

99 Prevention of Urinary Stone Disease

1 · STONE DISEASE AS A CHRONIC CONDITION

Recurrence is the disease, so prevention is lifelong.

- Up to **50%** recur within 5–10 years. The **ROKS** nomogram predicts a second stone.
- **Low urine volume** is the single most important risk; others are young age, male sex, obesity, family history and low citrate.

2 · SCREENING AND METABOLIC EVALUATION

P1 P2 *Metabolic Evaluation in Stone Disease — screening (history, blood, urine, stone analysis, imaging) for every first-time former; high risk: one or two 24-hour urines; fast/calcium-load testing discouraged*

Not in the paper answer:

- A 24-hour urine is **not** part of screening.
- Lithogenic drugs include calcium, vitamins C and D, acetazolamide and steroids.

3 · GENERAL MEASURES (ALL FORMERS)

- **Fluid** is the cornerstone: aim for urine **≥2.5 L/day**.
- **Limit nondairy animal protein and sodium**, which raise urinary calcium and lower citrate.
- Keep **calcium intake normal**, because restriction increases stones.

> **KEY POINT**
>
> Why normal calcium? Dietary calcium binds oxalate in the gut; restricting it frees oxalate for absorption.

4 · HYPERCALCIURIA: THIAZIDES

- The AUA makes thiazides the **preferred drug** for hypercalciuria with recurrent calcium stones, **and** for recurrent calcium stones with a normal evaluation.
- **Hypokalaemia causes intracellular acidosis, which leads to hypocitraturia**. So add **potassium citrate**, and check potassium 1–2 weeks after starting or changing the dose.
- With amiloride, **never add potassium**. Avoid triamterene, as it forms stones.
- Side effects include gout, diabetes and sexual dysfunction.

Target	Drug and dose	Key side effects
Calcium	Hydrochlorothiazide 25 mg twice daily; chlorthalidone 25–50 mg; indapamide 2.5 mg	Hypokalaemia, hyperuricosuria, hypocitraturia
Citrate	Potassium citrate 30–60 mEq/day	Gastrointestinal (GI) upset, hyperkalaemia
Uric acid	Allopurinol 300 mg	Rash, myalgia, hypersensitivity
Oxalate	Magnesium; pyridoxine 100 mg	Diarrhoea (magnesium); ataxia, numbness (pyridoxine)
Cystine	Tiopronin; D-penicillamine 250 mg titrated	See thiol table
Urease	Acetohydroxamic acid 250 mg three times daily	Heaviest burden (below)

5 · HYPEROXALURIA

- Stone risk rises continuously with urinary oxalate. Avoid large portions of spinach, nuts, chocolate and tea.
- **Enteric** hyperoxaluria follows malabsorption or Roux-en-Y gastric bypass; urine oxalate is >40 mg/day. Unabsorbed fat binds calcium, which frees oxalate for colonic absorption.
- Treat with **calcium up to 1 g/day with meals**, very high fluid, **liquid** potassium citrate, magnesium and a low-oxalate diet.

P2 *Primary Hyperoxaluria — oxalate often >**100 mg/day**; PH1 (~70%, AGXT, pyridoxine-responsive), PH2 (GRHPR), PH3 (HOGA1); lumasiran silences glycolate oxidase (ILLUMINATE-A)*

Not in the paper answer:

- Lumasiran is approved for PH1.

6 · HYPOCITRATURIA

- AUA offers **potassium citrate** 30–60 mEq/day for recurrent calcium stones with low citrate, **and** for those with no abnormality.
- Sodium citrate does not lower urinary calcium. Use caution in renal impairment.
- Over-alkalinising to **pH ≥7** creates **calcium phosphate** risk.
- **Alkali citrate, not total citrate, matters**.

7 · URIC ACID STONES: A PH DISEASE

P2 *Uric acid calculi: etiopathogenesis, medical management and allopurinol — pKa **5.5**; low pH the main cause; potassium citrate to **6.0–6.5**, avoid >7.0; allopurinol not first-line; hypersensitivity (fatal up to 27%)*

- Give allopurinol only for hyperuricaemia with hyperuricosuria, or failed alkalinisation.

8 · CYSTINE STONES

ALGORITHM

1. **Dilute**: urine ≥**3 L/day**, including at night; sodium ≤2300 mg/day; moderate protein (not in children).
2. Aim for cystine <**250 mg/L**.
3. If not enough, **alkalinise** with potassium citrate to pH **7.0–7.5**.
4. If still forming stones, add a **thiol** (tiopronin).

- pH >7.5 risks calcium phosphate, and sodium alkali raises cystine excretion.
- Thiols form a soluble cystine–drug disulfide.

	D-Penicillamine	Tiopronin	Captopril
Status	First thiol; largely replaced	**Thiol of choice**; fewer stop it	Weak evidence
Dose	250 mg/day, titrate	300 mg twice daily, up to 1200 mg/day	—
Toxicity	Rash, fever, arthralgia, pancytopenia, nephrotic syndrome	Asthenia, GI upset, rash, joint aches	—

9 · INFECTION (STRUVITE) STONES

- **Clear the stone completely**, because every fragment harbours bacteria.
- **Acetohydroxamic acid (AHA)** is a urease inhibitor. AUA: use it only for residual or recurrent struvite after surgical options are exhausted.
- AHA toxicity is ~30%, with **deep venous thrombosis** the most severe. It is contraindicated if creatinine is >**2.5 mg/dL** or in pregnancy.
- Most struvite formers are also hypercalciuric or hypocitraturic, so do a 24-hour urine.

10 · MONITORING

ALGORITHM

1. Start therapy.
2. Repeat the 24-hour urine within **6 months**.
3. Then check urine and blood (for side effects) at least yearly.
4. If a new stone forms, **re-analyse** it.
5. Image periodically.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Borghi 1996	Randomised controlled trial (RCT), high fluid, calcium oxalate formers	5-year recurrence 12% vs 27%	Urine >2 L/day prevents stones
Borghi 2002	RCT, hypercalciuric formers	Low protein and salt with normal calcium halved events vs low calcium	Never restrict calcium
Ettinger 1988	Double-blind RCT, chlorthalidone	~90% fewer stones	Thiazides prevent calcium stones
Barcelo 1993	RCT, potassium citrate	Fewer new stones at 3 years	Citrate prevents recurrence
ILLUMINATE-A	Phase 3 RCT, lumasiran, PH1	84% vs 0% reached near-normal urine oxalate	RNA interference for PH1

100 Lower Urinary Tract Calculi

1 · BLADDER CALCULI

- Bladder stones are ~5% of urinary stones. Types are **migrant**, **primary** (endemic) and **secondary**.

	Primary (endemic)	Secondary
Patient	Children; peak **2–4 years**; boys far more than girls	Men >60 with outlet obstruction

	Primary (endemic)	Secondary
Cause	Cereal diet low in animal protein and phosphate	Bladder outlet obstruction (BOO), foreign body, catheter, infection
Composition	Ammonium acid urate, calcium oxalate, uric acid	**Struvite**
Recurrence	Solitary; rarely recurs	Recurs unless the cause is removed

- Foreign bodies act as a nidus: a migrated intrauterine device, sutures, balloon fragments or hair during clean intermittent catheterisation (CIC).

> **KEY POINT**
> Primary stone is a disease of the urine (normal bladder); secondary is a disease of the bladder (stasis, nidus, infection). Do not hunt for outlet obstruction in the endemic child.

- Adults present with **terminal haematuria** and lower urinary tract symptoms.

2 · MANAGEMENT OF BLADDER STONES

The route, not the stone, dictates modality.

Option	Indication / key point
Chemodissolution	Adjunct only; for catheter encrustation
Extracorporeal shock wave lithotripsy (ESWL)	Artificial sphincter, penile prosthesis, neobladder, high-risk patient; works poorly in continent reservoirs
Transurethral cystolithotripsy	Most popular; holmium laser of choice; risks urethral stricture
Percutaneous cystolithotomy	Difficult urethra (bladder neck reconstruction or closure); spares the urethra
Open cystolithotomy	Small-capacity bladder, stricture, no safe endoscopic route

3 · SPECIAL SITUATIONS

Treat the cause along with the stone.

- **BOO** is present in 45–79%. **A bladder stone secondary to BOO is an absolute indication for BPH surgery**, but **prove BOO** first.
- **Diversion or augmentation** stones arise from acidosis-driven hypercalciuria, enteric hyperoxaluria, urease producers and **mucus**.
- A **continent pouch: go percutaneous**, which protects continence. A conduit with stomal stenosis needs stone clearance and stomal revision.
- **Spinal cord injury**: risk peaks **3 months** after injury; **CIC beats indwelling catheters**. Stones ≤4 cm get transurethral laser; larger ones percutaneous or open.
- **Transplant**: **remove all exposed suture**. **Children**: use percutaneous cystolithotripsy for large stones.

4 · URETHRAL CALCULI

- Urethral stones are 0.3–1% of stones, in males. Most are **migratory**, and the commonest site is the posterior urethra.
- They cause **acute painful retention**; anterior stones give a palpable mass. ~98% are radio-opaque.

Site	Treatment
Posterior	**Push back** into the bladder, then laser; else treat in situ
Anterior	**No pushback**; in situ holmium laser; urethrotomy for large stones

Site	Treatment
Fossa navicularis	Forceps; ventral meatotomy
Diverticular / stricture	Extraction with diverticulectomy or stricture repair, same sitting or staged

5 · UNUSUAL SITES

- **Preputial** stones follow phimosis; treat by circumcision and send the **excised skin to histopathology**.
- **Prostatic** stones are linked to BPH, **not cancer**.
- **Seminal vesicle** stones cause haematospermia; **TRUS is most accurate**.

54 Management of Paediatric Kidney Stone Disease

1 · EPIDEMIOLOGY

Children carry stone disease for life.

- Early onset means stones before **24 years**. About 50% recur within 3–5 years.

Feature	Children	Adults
Sex ratio	Female predominance	Male (historically 3 : 1)
Calcium oxalate (CaOx)	≈ **80%**, mostly idiopathic, on Randall's plaques	75–80%
Calcium phosphate (CaP) / uric acid	CaP commoner and uric acid rarer, because urine is more alkaline	≈ 5% each
Lifetime stakes	Chronic kidney disease (CKD), hypertension, low bone mineral density (BMD); cumulative radiation	Shorter

- **High-risk** groups include bladder augmentation (≈ 1%/year) and malabsorption.

> **KEY POINT**
> Childhood is the bone-accrual window, so seek low BMD early.

2 · DIAGNOSIS AND IMAGING

ALGORITHM

1. If a stone is suspected, do grey-scale **ultrasound (US)** with colour Doppler.
2. If US is negative but suspicion is high, or there is hydronephrosis, then do **low-dose non-contrast computed tomography (NCCT)**.
3. If US is negative and suspicion is low, stop.

- US shows the **twinkle artefact** (sensitivity > 70%, specificity > 95%) but overestimates size by 1–2 mm. NCCT is ≈ 100% accurate.

3 · METABOLIC EVALUATION: EVERY CHILD

An abnormality raises recurrence up to **fivefold**, so do a **full evaluation after the first stone**.

- **History**: ketogenic diet, vitamin D, furosemide, topiramate, prematurity.
- **Examination clues**: dysmorphism suggests Williams syndrome, rickets suggests Dent disease or HHRH, and tetany suggests FHHNC.
- Do **stone analysis** first: cystine, struvite, 2,8-dihydroxyadenine and xanthine are each diagnostic.
- With hypercalcaemia and hypercalciuria, check for **CYP24A1** loss.

Metabolite	Normal (child)
Calcium	< **4 mg/kg/day**; spot Ca : creatinine < 0.8 (0–6 mo), < 0.6 (7–12 mo), < **0.21** (> 2 y)
Oxalate	< 40–50 mg/1.73 m²/day
Citrate (lower limit)	Hypocitraturia is < 128 mg/g creatinine (male), < 300 (female)
Cystine	< 50–60 mg/1.73 m²/day
Uric acid	< 815 mg/1.73 m²/day; spot < 0.56 mg/dL glomerular filtration rate

KEY POINT

Never use adult limits: infants excrete more calcium, so the school-age 0.21 ratio overcalls them. Citrate's abnormality is too little.

4 · URINARY ABNORMALITIES

Hypercalciuria: commonest (30–50%)

Serum calcium	Causes
High	Hyperparathyroidism, CYP24A1, hypervitaminosis D, immobilisation, Williams
Low	Hypoparathyroidism; autosomal dominant hypocalcaemic hypercalciuria (calcium-sensing receptor gain)
Normal	Prematurity, loop diuretics, topiramate, ketogenic diet, Dent, Bartter, FHHNC, distal renal tubular acidosis (dRTA), HHRH; if none, **idiopathic**

Hyperoxaluria

P2 *Primary Hyperoxaluria — urinary glycolate/L-glycerate, then genetics; pyridoxine*

Not in the paper answer:

- Only 10–30% of primary hyperoxaluria type 1 (PH1) respond to pyridoxine.
- Thiols and lumasiran are the only drugs approved for paediatric stone prevention.

Others

- **Hypocitraturia**: incomplete dRTA needs an acid-load test. **Cystinuria** may present in infancy with a staghorn.

Monogenic disease	Inheritance	Signature
Dent (CLCN5)	X-linked	Low-molecular-weight proteinuria, nephrocalcinosis, CKD
Bartter	Recessive	Salt wasting, hypokalaemic **alkalosis**, hypercalciuria
FHHNC (CLDN16/19)	Recessive	Hypomagnesaemia, tetany, nephrocalcinosis
dRTA	Dominant/recessive	Hypokalaemic **acidosis**, hypocitraturia, nephrocalcinosis

5 · MANAGEMENT PRINCIPLES

- Tamsulosin is off-label (≤ 6 weeks).
- The residual fragment threshold is **4 mm on US**: 2 mm on CT plus the ≈ 2 mm US overestimate.
- Give antibiotic prophylaxis for all URS and PCNL, but for SWL only with infection risk; limit it to ≤ 24 h.

Scenario (AUA/Endourological Society 2016)	Recommendation
Uncomplicated ureteric ≤ **10 mm**	Observation with or without an α-blocker
Ureteric, failed or unlikely to pass	URS or SWL (strong)
Before URS	No routine pre-stenting
Renal ≤ **20 mm**	SWL or URS

Scenario (AUA/Endourological Society 2016)	Recommendation
Renal > 20 mm	PCNL or SWL (with stent or nephrostomy); authors prefer PCNL
Before PCNL	Low-dose CT (strong)
Asymptomatic non-obstructing renal	Surveillance with US
Open/laparoscopic/robotic	Only with anatomic abnormality

6 · URETEROSCOPY

The small ureter is the main limit, so never force access.

- Clearance is ≈ 90%, but worse under 6 years and in the lower pole.

ALGORITHM

1. Confirm a sterile culture; place in lithotomy; do cystoscopy.
2. Do a retrograde pyelogram and place a safety wire.
3. Calibrate the orifice with an 8/10 Fr coaxial dilator.
4. **If it will not pass, then place a 4.6–4.8 Fr stent for ≥ 14 days and return later.**
5. If it passes, do URS.
6. Stent if the case was long, needed many passes, or caused trauma or oedema.

- **No balloon or active orifice dilatation**; rely on passive stent dilatation.

7 · SHOCKWAVE LITHOTRIPSY

- SWL equals URS for small stones.
- Favourable: < 10 mm, non-lower-pole, < 1000 HU. Unfavourable: cystine, brushite, infundibulopelvic angle > 45°.
- Use < 20 kV and ≤ 2000–3000 shocks (< 2000 in infants); **80/min is better than 120/min**.
- Absolute contraindications are bleeding diathesis, untreated UTI and pregnancy.

8 · PERCUTANEOUS NEPHROLITHOTOMY

- PCNL is first-line for stones > 20 mm. A 24–30 Fr tract in a child equals ≈ 72 Fr in an adult, so **miniaturise**.

ALGORITHM

1. Prevent hypothermia; place a ureteric catheter.
2. Position prone (latex precautions in myelomeningocele).
3. Puncture a posterior calyx under US or fluoroscopy.
4. Dilate serially to an 18 Fr sheath (24 Fr for staghorn).
5. Fragment and check with a flexible scope.
6. Place a nephrostomy (7 Fr infant to 16 Fr adolescent).

9 · SECONDARY PREVENTION

- **Fluid**: at least maintenance; > 2–2.5 L/day in adolescents; 1.5–2 L/m² in cystinuria or PH.
- **Sodium** < 2–3 mEq/kg/day. **Calcium is not restricted**, because it binds gut oxalate.
- Target urine pH > 6.5 for uric acid and > 7 for cystine; avoid > 6.5 in calcium formers.

Drug	Indication	Child dose / caution
Hydrochlorothiazide	Hypercalciuria despite salt restriction	1–2 mg/kg/day; for hypokalaemia add potassium citrate

Drug	Indication	Child dose / caution
Potassium citrate	Hypocitraturia, uric acid, cystine	2–4 mEq/kg/day; CaP risk if pH > 6.5
Tiopronin	Cystinuria failing fluid and alkali	Fewer adverse effects than penicillamine
Allopurinol	HPRT, PRPS, adenine phosphoribosyltransferase deficiency; hyperuricosuric CaOx	4–10 mg/kg/day; can cause xanthine stones
Pyridoxine	PH1 (trial in all)	10–30% respond
Lumasiran	PH1	RNA interference against glycolate oxidase

PH1 with end-stage disease needs liver–kidney transplantation.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ILLUMINATE-A	Randomised, lumasiran vs placebo, PH1	Urinary oxalate lowered by 65%	RNA interference therapy for PH1

125 Genital and Lower Urinary Tract Trauma

1 · PRINCIPLES

- Key decision: **immediate reconstruction vs deliberate delay**. Testicular rupture is repaired at once; pelvic fracture urethral injury (PFUI) gets a suprapubic tube now and urethroplasty later.
- Imaging: ultrasonography for scrotum and equivocal penile fracture; computed tomography (CT) cystography for bladder; retrograde urethrography (RUG) for urethra.

2 · PENILE FRACTURE

P2 *Fracture penis — tunica thinnest ventrolaterally; snap, then detumescence, then eggplant deformity; deviates away from tear; urethral injury ~20% West vs ~3% Middle East; clinical diagnosis, ultrasound, then magnetic resonance imaging (MRI) Intraoperative flexible cystoscopy; delay ≤7 days harmless; complications ~20%*

Not in the paper answer:

- Bilateral tears (~10%) carry more urethral injury, and absence of blood at the meatus does not exclude it.
- Collagenase ruptures mostly occur after the 14-day observation window; manage them conservatively unless the urethra is injured.

ALGORITHM

1. Explore promptly. Through a ventral penoscrotal (or small lateral) incision, evacuate haematoma and find the tear (artificial erection).
2. Irrigate; close the tunica with interrupted **2-0/3-0 absorbable** sutures.
3. If the urethra is injured, repair it.

- A breached Buck's fascia gives a butterfly haematoma, and a negative MRI reliably excludes a fracture. Avoid deep corporal ligation and excessive debridement.

3 · PENETRATING, BITE, AMPUTATION, ZIPPER AND STRANGULATION

- **Gunshot**: explore immediately without waiting for imaging. Debride, give antibiotics and close the tunica.

- Urethral injury occurs in **15–50%** of penile gunshots, so check with RUG or intraoperative dye. Viable tissue is repaired primarily; high-velocity tissue loss needs suprapubic diversion and staged repair.
- **Dog bites**: irrigate, debride and close **primarily**, with amoxicillin–clavulanate and tetanus and rabies cover. **Human bites** present late and infected, so **leave them open** and give amoxicillin–clavulanate or moxifloxacin.

Penile amputation

- It is usually self-mutilation, so **always get psychiatric consultation**. **Microsurgical replantation** is preferred; otherwise repair urethra and corpora macroscopically.
- Transport the part **"double bagged"**: saline gauze in a sterile bag, inside an outer bag of ice slush.
- Erections return in >50% with either technique, but microvascular repair gives less stricture and sensory loss. If the part is lost, formalise the stump; total loss needs phalloplasty.

ALGORITHM

1. Suprapubic cystostomy; debride.
2. Tunica with 2-0 absorbable; urethra over a catheter with 4-0.
3. Anastomose dorsal artery with **11-0**, dorsal vein with **9-0** and dorsal nerve with **10-0** nylon.
4. Provide skin cover.

- **Zipper** injury (children): under penile block, try mineral oil once; if that fails, cut the slider's **median bar** with bone cutters.
- **Strangulation**: in children think of hair tourniquet and **child abuse**. Escalate from lubrication to the **string technique**, then to cutting tools with saline cooling in theatre.

4 · TESTICULAR INJURY

- About 80% of penetrating scrotal wounds also involve neighbouring structures, including the **femoral vessels**.
- Consider rupture in every blunt injury; haematoma size does not predict it. **Ultrasonography** sensitivity is only ~64–75%, so an **equivocal scan must not delay exploration**.
- A non-palpable testis after trauma suggests **dislocation**, commonest to the superficial inguinal pouch (50%). Trauma does not exclude torsion.

AAST testis grade	Injury
I	Contusion/haematoma
II	Subclinical laceration of tunica albuginea
III	Tunical laceration with <50% parenchymal loss
IV	Major laceration with ≥50% parenchymal loss
V	Total destruction or avulsion

ALGORITHM

1. Explore; remove necrotic and extruded tubules.
2. Close the tunica with small absorbable sutures.
3. If the defect is large, use a tunica vaginalis flap or graft (never synthetic); drain.

- Explore a **significant haematocele** (up to 80% hide rupture) and an **intratesticular haematoma** (delayed orchidectomy 15%).
- Always explore penetrating injury; ligate an injured vas and reconstruct later.
- Salvage is **>90%** if repaired within **72 h**; non-operative care leads to infection, atrophy and late orchidectomy.

5 · GENITAL SKIN LOSS

- Fournier gangrene debridement is the commonest cause. Treat it with antibiotics, resuscitation and **repeated debridement**, then negative pressure wound therapy; hyperbaric oxygen is unproven.

	Penis	**Scrotum**
Graft	**Thick, non-meshed** split-thickness skin graft (STSG) 0.012–0.015 inch, because meshed graft contracts	**Meshed** STSG after local flaps (exudate escapes, so better take)
Key step	Excise residual subcoronal skin, which otherwise causes lymphoedema; never reuse avulsed skin	Suture the testes together to avoid a bifid scrotum; ≤50% loss closes primarily
Note	Graft is insensate, but sexual function is preserved	Thigh pouches not recommended

- Graft 5–7 days after the last debridement, once the wound is granulating, under a bolster or vacuum dressing.

6 · BLADDER INJURY

Intraperitoneal tears need surgery; most extraperitoneal tears heal on a catheter.

P2 *Lower urinary tract trauma in pelvic fracture: classification, assessment and (early) management of bladder and urethral injuries — pelvic fracture in 83–95% of bladder injuries, bladder in 5–10% of fractures; cystography for gross haematuria and pelvic fracture; retrograde fill* ***350 mL*** *, drainage film (plain only); flame-shaped extraperitoneal; 22-Fr catheter for uncomplicated extraperitoneal; intraperitoneal needs immediate repair; suprapubic tube adds nothing*

Not in the paper answer:

- Underfilling the bladder gives false-negative cystograms.
- Bone spicules in the wall prevent healing, so repair them; a ureteric orifice injury needs a stented reimplant.

AAST bladder grade	**Injury**
I	Contusion, intramural haematoma, partial-thickness laceration
II	Extraperitoneal laceration <2 cm
III	Extraperitoneal ≥2 cm, or intraperitoneal <2 cm
IV	Intraperitoneal laceration ≥2 cm
V	Laceration extending into bladder neck or ureteric orifice (trigone)

- Intraperitoneal leakage causes chemical peritonitis and, once reabsorbed, **acidosis and azotaemia**.

> **KEY POINT**
> Immediate repair: intraperitoneal, penetrating, bladder neck, rectal/vaginal injury, bone fragments in bladder, clots/poor drainage, iatrogenic urological injury, laparotomy for other reasons.

7 · POSTERIOR URETHRAL INJURY (PFUI)

Drain now and reconstruct later.

P2 P3 *Lower urinary tract trauma in pelvic fracture / early and late management of PFUD — bulbomembranous junction; triad blood at meatus, retention, palpable bladder; RUG technique; suprapubic tube standard, brief realignment optional (no difference); delayed excision and primary anastomosis at* ***2–3 months*** *(corporal split, then pubectomy, then rerouting); erectile dysfunction (ED) up to 82%, restenosis 5–15%, incontinence <4%*

Not in the paper answer:

- PFUI complicates ~10% of male and up to 6% of female pelvic fractures, mostly **straddle (four-rami)** and unstable fractures.

- ED is predicted by symphyseal diastasis, lateral displacement and a long gap. Endoscopic "cut-to-the-light" fails for gaps ≥1 cm.

AAST urethra grade	Injury
I	Contusion — blood at meatus, normal urethrogram
II	Stretch — elongation without extravasation
III	Partial disruption — extravasation, contrast reaches bladder
IV	Complete disruption, <2 cm separation
V	Complete, >2 cm separation or extension into prostate/vagina

ALGORITHM

1. If there is blood at the meatus, do RUG before any catheter.
2. Place a 16-Fr catheter in the fossa navicularis and inflate the balloon with 3 mL.
3. Apply penile traction, position oblique, inject 20 mL contrast and image during injection.

- In females, examine the vagina in every pelvic fracture. A suprapubic tube is safe with pelvic hardware if tunnelled high in the midline.

8 · ANTERIOR URETHRAL INJURY

- Most are **straddle injuries** crushing the bulbar urethra against the pubis. Contusion or incomplete tear needs a catheter; major injury a suprapubic tube.
- Later, the 1.5–2 cm obliterated bulbar segment is treated by **excision and primary anastomosis** (>95% success). Urethrotomy through an obliteration is futile.

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Multi-institutional PFUI study	Prospective, suprapubic tube vs early endoscopic realignment	No difference in stenosis or urethroplasty need	Suprapubic tube remains standard; realignment optional, brief

62 Paediatric Genitourinary Trauma

1 · RENAL INJURY: HOW THE CHILD DIFFERS

- The kidney is the commonest site of paediatric genitourinary (GU) trauma. Most injuries are **blunt** and grade 1–3; half have other injuries.
- Congenital anomalies such as ureteropelvic junction (UPJ) obstruction are commoner than in adults.

Feature	Child	Adult
Protection	Less perirenal fat, weaker wall, kidney below the ribs	Better cushioned
Blood pressure	Maintained until ≈ **25–30% volume loss** ; hypotension is late	Falls earlier
Haematuria	May be absent (up to 2/3 of grade ≥ 2 injuries)	Classic trigger
Imaging decision	**Mechanism-led**; keep radiation as low as reasonably achievable (ALARA)	Led by haematuria and hypotension

2 · INDICATIONS FOR IMAGING

Setting	Indication (American Urological Association [AUA])
Stable blunt	Gross haematuria
Stable blunt	Microscopic haematuria with systolic blood pressure (SBP) < 90 mm Hg
Mechanism / examination	Rapid deceleration; significant flank blow; rib fracture; flank ecchymosis
Penetrating	Any injury to abdomen, flank or lower chest, even without haematuria

- Triggers: fall > **10 feet (3 m)**, high-speed collision, flank blow; rib, spine, femur or pelvic fracture. All four markers together (haematuria, hypotension, associated injury, mechanism) give > **98% sensitivity**.
- Paediatric hypotension (1–10 years) is SBP < 70 + (2 × age) mm Hg.

KEY POINT

Trap: "no haematuria, so no imaging". In children both adult markers fail, so mechanism drives the decision.

3 · IMAGING STUDIES

- Renal ultrasound (US) detects grade 3–5 without radiation and suits screening, but **contrast computed tomography (CT)** is preferred. Add a **delayed (excretory) phase** if a leak is suspected.

4 · GRADING

American Association for the Surgery of Trauma (AAST): grades 1–3 are **low** and 4–5 are **high**.

Grade	1989 AAST	2011 revision
1	Contusion or non-expanding subcapsular haematoma	Subcapsular haematoma/contusion
2	Perirenal haematoma; cortical laceration < 1 cm	Same, within Gerota fascia
3	Laceration > 1 cm, no collecting-system injury	Same, into medulla
4	Laceration into collecting system; main artery/vein injury with contained haemorrhage	Collecting system with extravasation; segmental vessel injury; pelvis laceration or **UPJ disruption**; shattered kidney
5	**Shattered kidney**; hilar avulsion	Main artery/vein laceration, avulsion or thrombosis

KEY POINT

Grades 1–3 differ only in depth (1 cm) and never involve the collecting system; entry makes grade 4. Shattered kidney is grade 5 (1989) but grade 4 (2011).

High-risk CT criteria (grade 4–5)

Criterion	Predicts
Perirenal haematoma > 3.5 cm adult, > **2.2 cm child**	Bleeding needing angioembolisation
Vascular contrast extravasation; medial laceration	Bleeding
> 25% parenchyma devitalised	Bleeding and open surgery
Medial urinary extravasation	UPJ disruption
Extravasation with no contrast in distal ureter	Urine leak needing intervention

With 0–1 factor, < 10% need intervention; with 2–3, > 50% do (grade 4a vs 4b).

5 · NON-OPERATIVE MANAGEMENT (NOM)

- NOM is the **default at every grade**, including grade 4–5 and stable penetrating injury, provided the child is stable.

ALGORITHM
1. Confirm stability; grade the injury.
2. Bed rest, with serial haemoglobin and abdominal examination.
3. When gross haematuria resolves and counts are stable, mobilise.

- Grade 1–2 needs only **23-hour observation**. Repeat imaging: AUA CT at 48 h (grade 4–5), EAU US at 2–3 days; but **symptoms, not imaging, predict complications**.
- Grade 3 (devitalised) to 5: ≈ **25%** need embolisation, ≈ **15%** urinoma intervention, ≈ 5% exploration.

6 · COMPLICATIONS DURING NOM

- **Urine leaks** occur only in grade 4–5, and up to **85%** resolve. A urinoma shows **flank pain, ileus and low-grade fever**.
- If no contrast reaches the distal ureter, do retrograde pyelography (RPG). Escalate from **stent and Foley catheter** (AUA first line, ≈ 90%; out at 4–6 weeks) to percutaneous drain, then nephrostomy, then open repair.

Bleeding	Persistent	Delayed
Mechanism	Vessels fail to tamponade	Fistula or pseudoaneurysm ruptures as the haematoma liquefies
Timing	From injury	Typically **1–2 weeks** (up to 1 month)
Stable	Allow 2–3 transfusions, then embolise	Early embolisation unless mild
Unstable	Urgent angioembolisation	

- Superselective embolisation succeeds in > 80%.

7 · OPERATIVE MANAGEMENT

- Operate for **instability from renal bleeding**, pulsatile or expanding haematoma, or failed embolisation. A stable non-pulsatile haematoma found at laparotomy is **left alone**.
- A **one-shot pyelogram** (2 mL/kg, film at 10–15 min) confirms a **contralateral kidney**.

P2 *Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management — vascular control, renorrhaphy, nephrectomy; revascularise only solitary/bilateral; Page kidney*

Not in the paper answer:
- Early vascular control did not lower nephrectomy in the one randomised trial.

ALGORITHM
1. Confirm a contralateral kidney.
2. Open Gerota fascia; control the vessels; debride; close the collecting system, then parenchyma over a bolster.
3. Drain. If the injury is penetrating, inspect the ureter.

- **UPJ disruption** follows deceleration, often without haematuria; CT shows **medial extravasation, empty distal ureter and often no laceration**. Repair early if stable; otherwise nephrostomy and repair at 2–3 months.

8 · FOLLOW-UP

- Scarring: grade 1–2 none, grade 3 ≈ 50–60%, grade 4–5 ≈ 100%, so do **US at 2–4 months**.
- **Hypertension** is rare and renin-mediated, so use renin–angiotensin blockade. Check blood pressure yearly after grade 3–5.

9 · URETERAL TRAUMA

- Most are penetrating. **Haematuria is absent in 25–50%** and ≈ 10% are missed at laparotomy, so diagnose by **delayed-phase CT** or RPG.

AAST grade	Injury
I	Contusion/haematoma, no devascularisation
II / III	< 50% / ≥ **50%** transection
IV	Complete transection, < 2 cm devascularisation
V	Avulsion, > **2 cm** devascularisation

- Found within 1 week: repair if stable; ligate and place a nephrostomy if unstable. Found after 5–7 days: stent first; if that fails, nephrostomy and reconstruction at 6–12 weeks.

P3 *Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury — repair by level: ureteroureterostomy, Boari, reimplant ± psoas hitch; ileal ureter, autotransplant*

Not in the paper answer:

- **Avoid transureteroureterostomy in children**, because a long life puts the other kidney at risk.

10 · BLADDER INJURY

Feature	Child	Adult
Bladder position	**More intraperitoneal**, so isolated injury is commoner (≈ half have no fracture)	Pelvic organ
Pelvic fracture	Half as frequent, so it implies higher energy and more urethral injury	Commoner
Posterior urethral injury	Prostatic; **bladder neck injury commoner**	Bulbomembranous

- **Gross haematuria or retention** calls for **retrograde cystography** with retrograde urethrography (RUG); delayed-phase CT may miss a rupture.
- AAST: I contusion; II extraperitoneal < 2 cm; III extraperitoneal ≥ 2 cm or intraperitoneal < 2 cm; IV intraperitoneal ≥ 2 cm; V bladder neck or trigone.
- Extraperitoneal tears get a catheter, but are repaired for bone fragments, hardware, rectal or vaginal tear, or bladder neck injury. Intraperitoneal tears are **always repaired**. Keep the catheter 1–2 weeks, then do a cystogram.

> **KEY POINT**
> Bladder neck avulsion is commoner in children; girls depend more on the bladder neck for continence, so outcome is poor and many need secondary procedures.

11 · URETHRAL INJURY

Grade/type	AAST	Goldman
I	Contusion: meatal blood, normal RUG	Posterior urethra stretched, intact
II	Stretch, no extravasation	Rupture above intact urogenital diaphragm
III	Partial: extravasation, contrast reaches bladder	Rupture through diaphragm into bulbar urethra
IV	Complete, no bladder filling, < 2 cm gap	Bladder rupture extending into proximal urethra

Grade/type	AAST	Goldman
V	≥ 2 cm gap or into prostate/vagina	Anterior urethral injury

- Do **RUG before any catheter**; if a catheter will not pass, place a **suprapubic tube**.
- Realignment is a brief attempt only; EAU advises against immediate urethroplasty. **Bladder neck injury is repaired at once.**

ALGORITHM

1. Divert with a suprapubic tube and wait several months.
2. Assess with cystogram and RUG, with or without endoscopy.
3. Do transperineal excision and primary anastomosis; if there is tension, add pubectomy.
4. Catheter for 2–3 weeks; cystogram; remove the suprapubic tube.

- Restricture ≈ 10%, incontinence ≤ 20%, and erectile dysfunction up to half (gap > 2.5 cm).

12 · GENITAL INJURY

KEY POINT

Most are accidental, but **abuse must always be assessed** (≈ 1 in 7 boys, 1 in 3 girls abused). Involve social services before discharge.

- Penis AAST: I skin; II Buck fascia; III avulsion, glans/meatus laceration, defect < 2 cm; IV partial penectomy or ≥ 2 cm; V total penectomy.
- **Circumcision** is the commonest cause; the **Mogen clamp** causes the worst (glans amputation).
- Testis AAST: I contusion; II subclinical tunica tear; III < 50% loss; IV ≥ 50% loss; V destruction. If US shows tunica discontinuity, haematocele or focal flow loss, **explore and close the tunica**.
- Hymenal, vaginal or perianal injury means **penetration: evaluate for abuse**; a **normal examination does not exclude abuse**.

★ Putting it together

A · RENAL COLIC

ALGORITHM

1. Give a non-steroidal anti-inflammatory drug (NSAID; ketorolac 30 mg) unless there is renal failure, bleeding or pregnancy. Confirm with non-contrast CT, or ultrasound in pregnancy and children.
2. With infection, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration or refractory pain, **decompress now** (stent or nephrostomy; equal). Never lithotripsy while infected.
3. **Otherwise**, if the stone is <10 mm, try passage with or without an α-blocker (distal 5–10 mm) for 4–6 weeks; if that fails, do ureteroscopy (URS) or shock wave lithotripsy (SWL).

B · STONE OPERATION

Stone	First choice	Notes
Ureteric, any level	URS or SWL	URS clears more, SWL has fewer complications; hard proximal >15–20 mm: consider percutaneous nephrolithotomy (PCNL)
Renal <2 cm, not lower pole	SWL or URS	SWL works best <900 Hounsfield units and skin-to-stone <10 cm
Lower pole >1 cm	PCNL or URS	SWL not first line (Lower Pole I)

Stone	First choice	Notes
Burden >2 cm, staghorn	PCNL	Staged URS an option

KEY POINT

SWL absolute contraindications: pregnancy, coagulopathy or antithrombotics, untreated infection, uncontrolled hypertension, arterial aneurysm, and obstruction distal to the stone. No routine stent after SWL or uncomplicated URS.

C · PREVENTION

ALGORITHM

1. **All patients:** urine ≥2.5 L/day, normal calcium, low sodium and less animal protein.
2. **If high risk**, give therapy guided by 24-hour urines, repeated within 6 months.

Abnormality	Treatment
Hypercalciuria	Thiazide and potassium citrate (hypokalaemia lowers citrate)
Hypocitraturia	Potassium citrate
Uric acid	Alkalinise to pH 6.5–7; allopurinol only for hyperuricosuria/gout or failed alkali
Cystine	≥3 L/day fluid, pH 7.0–7.5, then tiopronin
Struvite	Complete clearance; check metabolism

D · RENAL TRAUMA

ALGORITHM

1. **Image** with delayed-phase CT for gross haematuria, microhaematuria with shock, deceleration or penetrating injury.
2. If stable with grade I–IV (or selected V), manage **non-operatively** (>90% succeed). Embolise a contrast blush; stent a symptomatic leak.
3. If unstable or expanding, **explore**; a one-shot pyelogram (2 mL/kg, 10 min) confirms the other kidney.

E · BLADDER AND URETHRA

Injury	Management
Extraperitoneal bladder rupture	22 Fr catheter, cystogram at 2–3 weeks; repair if at laparotomy, bone spicule, or unhealed at 4 weeks
Intraperitoneal bladder rupture	Surgical repair
Pelvic fracture urethral injury	Suprapubic tube (or brief realignment) now, delayed urethroplasty later
Penile fracture	Prompt tunical repair; check the urethra
Testicular rupture	Immediate repair

Retrograde cystography must fill to ≥350 mL.

F · URETERIC DEFECTS

Defect	Repair
2–3 cm, upper/mid	Spatulated ureteroureterostomy

Defect	Repair
Lower ureter, 6–10 cm	Psoas hitch reimplant
10–15 cm	Boari flap
Longer	Transureteroureterostomy, ileal ureter (creatinine <2 mg/dL), autotransplant, nephrectomy

G · CHILDREN

KEY POINT

Stones: ultrasound first; full metabolic evaluation after one stone. Renal ≤20 mm: SWL or URS; >20 mm: PCNL. **Trauma:** image by mechanism (fall >3 m, deceleration); a perirenal haematoma >2.2 cm predicts embolisation.